Microsoft Windows Server

OSSE1091E

Course Module 1

Book Module 1 and 2

Introduction to Virtualization

What is Virtualization?

- Classically Defined: Running two or more operating systems on one instance of physical hardware.
- More than one type of Virtualization
  - Desktop virtualization
  - Application Virtualization (App-V)
  - Storage Virtualization (SAN)
  - Server (Hardware) Virtualization
  - Network Virtualization (SDN)

Why do we use virtualization?

- Consolidation of server(s)/client(s)
- Efficient use of Hardware/Cost effectiveness
- Multi-OS on one physical host machine.
- Foundations of Cloud computing

Disadvantages of Virtualization?

- Single point of failure.
  - Requires hardware redundancy to mitigate.
- Overprovisioning of a single resource can lead to bottlenecks.
  - Example: backing up 10 VMs *could* be slower than backing up several physical servers (over the network) due to the VMs using the same network connection/HDD/CPU.

Anatomy of a VM

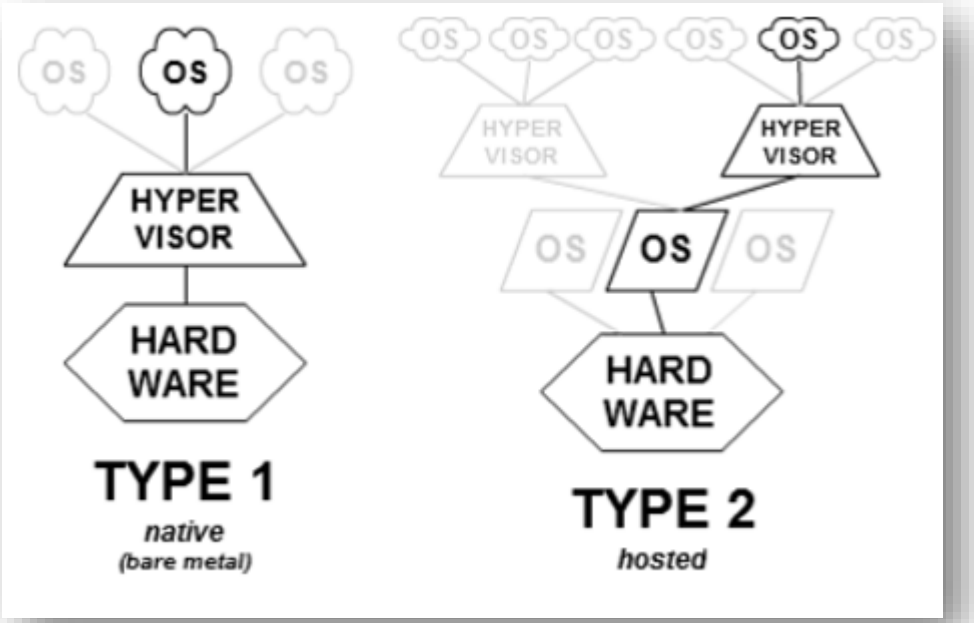
Hypervisor

- A Hypervisor.
  - The Management software for Virtualization
  - This “supervisor” of the “supervisor”.

- Other Hypervisors
  - Hyper-V (Windows)
  - ESX/ESX-I (VMware, enterprise)
  - VMWare Fusion
  - Bootcamp
  - Parallels
  - Virtual Box (Oracle)

Hypervisor

- VMWare Player, and Hyper-V are Hypervisors.
  - The Management software for Virtualization
  - This “supervisor” of the “supervisor”.
- Native (Hyper-V, ESXi)
- Hosted (VirtualBox, VMWare Player, Parallels)



https://en.wikipedia.org/wiki/Hypervisor

Guest vs. Host?

- The HOST machine is the physical (sometimes Virtual) computer that runs the Hypervisor.
- The GUEST machine is the VM.
  - Speaks *through,* and *to*, the HOST by way of the hypervisor.

How Virtualization works (the short version)

- Everything is just software.
  - Network card? Just a bunch of settings in a file.
  - Hard Disk? Just a really big file called a VMDK/VHDX.
  - Sound and Video? All piggy backing on the Host Machine.
- This is all possible through a process called Hardware Abstraction
  - "[…] sets of routines in software that provide programs with access to hardware resources through programming interfaces."
  - Hypervisor acts a middleman between the VMs and the hardware of a Host machine.
  - Every resource used by the Guest is taken from the host.

https://en.wikipedia.org/wiki/Hardware_abstraction

Virtual Disks

- Large files that represent Physical Disk Structures
  - Can be split into multiple smaller files (to facilitate moving the files easier).
- For VMWare the file is called a VMDK
- VMDKs are Dynamically expanding (they are only as large as what's in them). Windows has a Native V-Disk format: VHD/X.

Virtual Disks

- Fixed VS. Dynamic (Thin/Thick Provisioning)
  - Fixed: the vDisk is exactly this size - always.
  - Dynamic: the disk is only as big as the files created within the vDisk.

Why are we using VMWare?

- Alternative is Hyper-V
- Hyper-V is free but can only be installed on Windows Professional/Enterprise/Educational.

Things to remember.

- The VM is ***blissfully unaware*** that it is just set of files. It will behave exactly like a physical computer, and other computers on the network will treat it as a physical computer.
- VMs ***matter***! Just because the machine isn’t real, doesn’t mean the DATA isn’t. VMs are just as important to protect as all other computers.
- If running local VMs (on your laptop) shutdown or suspend your VM when you're not using it (it WILL continue to take resources).

NBCC Academic Computer Center (ACE)

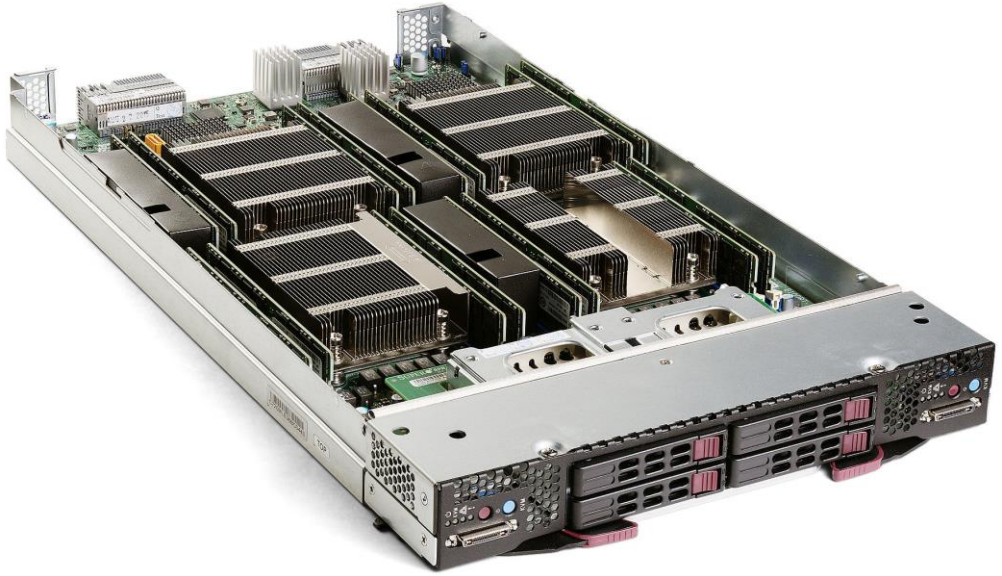
- NBCCs own personal cloud (think MS Azure, or Amazon AWS).
- Requires the use of a site-to-site VPN.
- Is much faster than your laptops
- Can be access from anywhere there is internet.

Servers: Physical VS. Virtual Concepts

Physical Servers

- Can be a tower (ATX/BTX) or other Desktop form factor, a rack-mounted formfactor, or a blade (high density) formfactor (there are others).
- Rack-mounted Servers are mounted in a server rack and usually share a single screen and mouse/keyboard using a Keyboard/Video/Mouse (KVM switch).
  - Although a server rack contains a KVM, it is more common to access these devices using remote technology like Remote Desktop Protocol (RDP).
- The Racks are partitioned into "U"s of measured space
  - 1 U is 1.75 inches (~4.5 cm) in height
  - The space that a server consumes is measure in U's
- Some rack-mounted servers can also be comprised of removable modules called blades.

Tower and Rack-Mounted Server form factor



https://newserverlife.com/articles/form-factors-of-server-equipment-purpose-of-using-types-and-features/

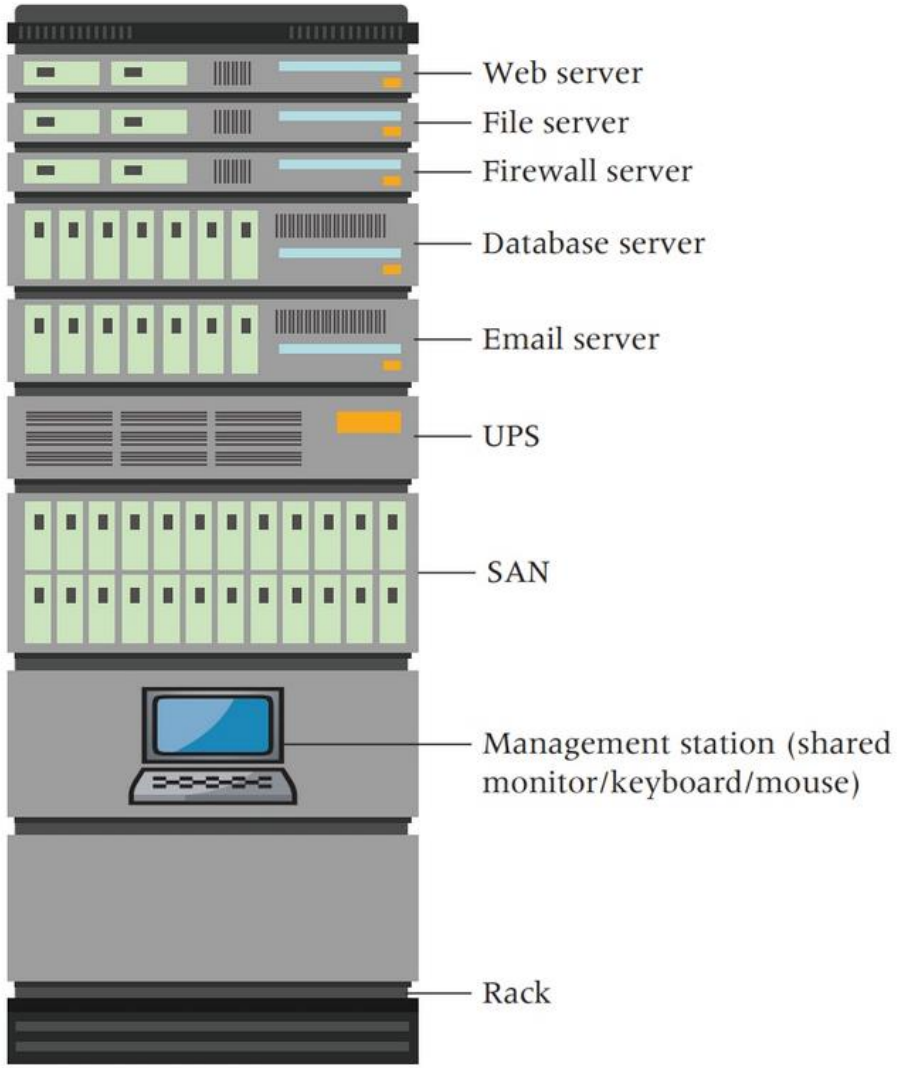
By Dmitry Nosachev - Own work, CC BY-SA 4.0, https://commons.wikimedia.org/w/index.php?curid=46899884

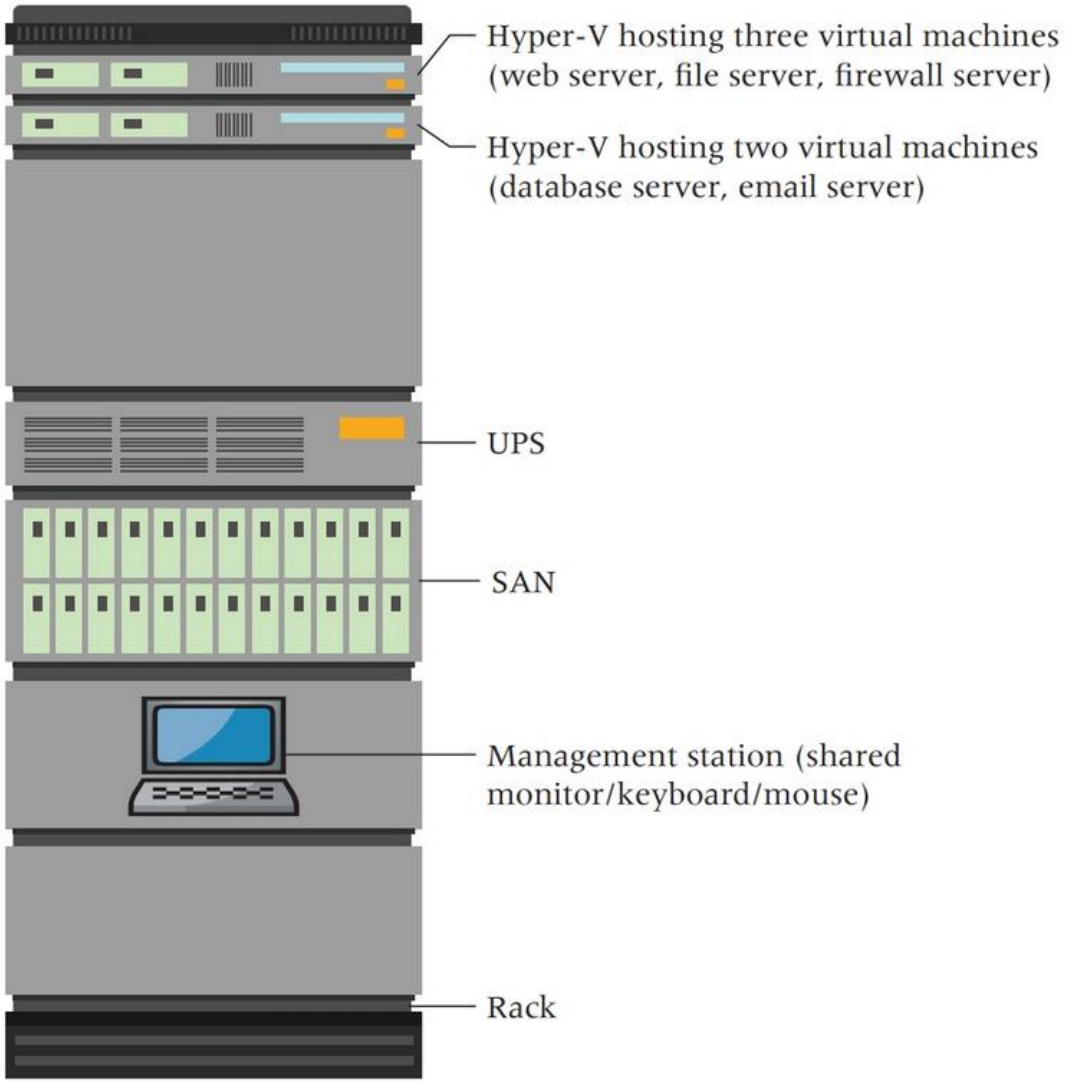
Blade Server Form factor

Blade Server
- Modular
- Compact

Blade Enclosure
- Contains one or more Blade Modules

Server Rack Examples (Physical/Virtual)





Virtualization

- Read: Book Module 1 (pages 3-11) for an explanation on more advanced Virtualization configurations.

Introduction to Windows Server

Why we need Server and what are they.

- Since the 1990's computers networks have been the primary method of distributing data within an organization.
- A network is composed of two *main* objects:
  - Servers: *Share* resources on a network
  - Clients: *Connect* to resources on a network
- While the operating system does not *technically* define what a server is Microsoft Windows has a product that is dedicated to, and created specifically for, sharing resources on a network: Windows Server.
- Currently, Windows Server is found in ~70% of all organizations, worldwide (as of 2019).

Why we need Server and what are they.

- Currently servers can be divided into two sub-groups:
  - On-premises (or ‘on-prem’): Severs that reside *within* an organizations local network.
  - Cloud: Servers that exists in data centers *outside* the organizations local network.

Clients + Servers = Domains?

- The client workstation operating system most compatible with Windows Server 2019/2022:
  - Windows versions 7, 8, 8.1, and 10/11
  - Windows 10/11 being the most compatible in terms of client management
- Client
  - A computer that accesses resources on another computer via a network
- Workstation
  - A computer that has its own central processing unit (CPU) and can be used as a stand-alone or network computer
- Server
  - Host services to be accessed by clients VIA a network or direct connection.

Clients + Servers = Domains?

- Domain (Active Directory (AD) Domain)
  - A grouping of network objects, such as computers, servers, and user accounts, that provides for easier management.
  - Computers and users in a domain can be managed (centrally) to determine what resources they can access.
- Domains have a unique name.
  - DNS domains are for finding resources.
  - AD domains are for organizing resources.
  - AD and DNS domains work **together**.

(Clients + Servers = Domains) + Active Directory?

- Active Directory
  - Database of computers, users, groups of users, shared printers, shared folders, and other network resources
  - Hierarchical management of network elements including authentication, authorization, access permissions, rights, privileges, polices, physical assets, human assets (users), and more.

Much more on this in Chapter 4

Peer-to-peer VS Server-Based Networking

Peer-to-peer

- Spreading out resources among all server and non-server members of the network.
- Physical Workstations and servers are both logical clients and servers.
- Effective only in small networks (<10 clients/servers total)

Server-Based (Domain-Based)

- Centralizes administration on one or more servers.
- Centralize access and control of network resources; monitoring.
- Servers are single computers with multi-user access. Physical workstations are (usually) clients only.

Server Features

Server Key Features

- Active Directory
  - Microsoft's flagship software for organizing and securing a Domain network.
  - Can also be deployed in the cloud using Azure Active Directory
- Security
- Volume and filesystem features
- Performance and reliability
- Administration tools
  - Server Manager: manage one or many Servers in simultaneously from a single console.
  - Windows Admin Center: A web-based server addon to do what Server manage does from the Web
- Small footprint installation options
  - Server Core: a minimalized installation of sever that includes only the required components of Windows Server.
  - Server Nano: Ideal for cloud; runs *only* essential services (DNS, DHCP, etc).

Windows Server 2019/2022 Editions

- Windows Server 2019/2022 platforms:
  - Windows Server 2019/2022 Essentials Edition
  - Windows Server 2019/2022 Standard Edition
  - Windows Server 2019/2022 Datacenter Edition
- Additional Windows Server 2019/2022 platforms targeted for specific-purpose niches:
  - Windows Storage Server 2019/2022
  - Windows Hyper-V Server 2019/2022

Windows Server 2019/2022 Essentials Edition

- Windows Server 2019/2022 Essentials Edition:
  - 25 users
  - Per processor licensing
  - 64 GB RAM
  - 16.8 million connections for file sharing through Server Message Block (SMB) services
  - 2 central processor sockets
  - 50 Remote Desktop/Remote Access connections
  - 50 Routing and Remote Access connections
- In addition:
  - This edition cannot join a domain, other than to migrate files and data from one server to another. It can *host* a domain with a single Domain Controller.
  - Provides most, but not all server roles (cannot hold the Hyper-V role or Host Cloud services), but it can *be* a Virtual Machine.
  - Only Available with OEM purchases.

Windows Server 2019/2022 Standard Edition

- Windows Server 2019 Standard Edition:
  - Unlimited users (user license limited).
  - Per core licensing
  - 24 TB RAM
  - 16.8 million connections for file sharing through Server Message Block (SMB) services
  - 64 central processor sockets
  - No limit on Remote Desktop/Remote Access connections
  - 2 VM client limit
- Standard Edition provides:
  - Access to all Server Roles
  - Hyper-V which enables servers to offer a virtualization environment (up to **two** clients)
  - Virtualized Containers
  - Clustering
  - Centralized management of applications and network resources

Windows Server 2019/2022 Datacenter Edition

- Windows Server 2019 Datacenter Edition supports a maximum of:
  - Unlimited users (user license limited).
  - Per core licensing
  - 24 TB RAM
  - 16.8 million connections for file sharing through Server Message Block (SMB) services
  - 64 central processor sockets
  - No limit on Remote Desktop/Remote Access connections
  - Unlimited VM clients
- Designed for environments with:
  - Mission-critical applications
  - Very large databases
  - Very large virtualization requirements
  - Cloud computing needs
  - Information access requiring high availability
- Offers support for clustering up to 64 computers

Windows Server 2019/2022 Datacenter Edition

- Differences between Standard Edition and Datacenter Edition: Datacenter Edition focuses on Datacenter Edition's industrial strength capabilities in the areas of:
  - Virtualization
  - Cloud computing
  - Database handling
- Datacenter is designed to provide the OS resources to accommodate large database applications

Edition Comparison

Table 1-1 Comparing the major Windows Server 2019 editions

Limitations	Essentials Edition	Standard Edition	Datacenter Edition
Maximum users	25	Limited by number of CALs	Limited by number of CALs
Licensing	Per server	Per processor core	Per processor core
Maximum memory	64 GB	24 TB	24 TB
Maximum processor sockets	2	64	64
Can join a domain	Only to enable migration	Yes	Yes
Maximum file sharing connections	16.8 million	16.8 million	16.8 million
Maximum remote access connections	50	Unlimited	Unlimited
Hyper-V	No	Yes (2 virtual machines)	Yes (unlimited virtual machines)
Containers	No	Yes (unlimited)	Yes (unlimited)
Hyper-V containers	No	Yes (up to 2)	Yes (unlimited)

Windows Server Core

- Windows Server Core
  - A minimum server configuration
  - Designed to function in a fashion similar to traditional UNIX and Linux servers
  - Does not provide the following:
    - A graphical interface, just a command line
    - Graphical tools to configure the server
    - Extra roles/features/services that you do not need
    - other software that relies on having GUI

Sever Core. Why server core?

- Smaller “footprint” (less resource intensive)
  - Uses 32 GB of disk space
  - 512MB RAM
  - 1GHz Proc
  - Minimal running services, Roles and features.
- Smaller attack surface.
  - The is no GUI. The GUI is often the target of Windows Exploits that can compromise a workstation.
  - No GUI; No GUI Exploits.

Installing Windows Server Core

- The steps for installing Windows Server Core are nearly identical to the steps for a full installation you simply do not choose "Desktop Experience" when installing Server.
- You can implement most server roles
- For some of these roles, not all role services can be installed as in the full installation of Windows Server 2019/2022
- Use `sconfig` CMD command for initial configuration of the server.
- To access PowerShell, type `powershell` at the command prompt in the CMD

More Later…

General Minimum H/W req. for Server 2019/2022

- CPU: 1.4Ghz 64-bit Processor (must be Virtualization capable Intel VT, AMD-V, SLAT)
- RAM: 512 MB (CORE)/2 GB (GUI) (ECC RAM Recommended)
- HDD: 32 GB (CORE)/ 36 GB (Desktop Experience)
- NIC: PXE-capable 1 Gbit*
- Trusted Platform Module (TPM) chip if using Secure boot.

*doesn’t have to be Gbit, but good luck on find a new server that isn't.

Initial Setup and Configuration

Post-Installation

Installation of Server

- Read: Book Module 1 (page 30-38) for a refresh on installing Windows Server.

Determining Whether to install Server Core, or the Full GUI Version

Core?

- The server is dedicated to one narrow task
- The server has low resources available, or the function of the server will not require many resources.

GUI (Desktop Experience)?

- You prefer to work in a GUI environment
- Your organization needs to have GUI-based software on the server
- The server administrator (or their delegate) is relatively new to Windows Server 2019/2022 and wants to use wizards for guidance (or cannot use PowerShell)

User/Device Licensing (CALs)

- Remote Desktop CALs
  - Remote Desktop Services (RDS, formerly Terminal Services is a Role that allows for multiple simultaneous (or concurrent) users to be able to remotely login to a single server and work in that environment.
  - One CAL must be purchased for each version (2019, 2022) of Server.
  - licensed separately on a per User or Device basis on top of the Server Core and Server CAL licensing. A user connecting remotely will require both the base CAL (user/device) *and* an RDS CAL.

Determining Domain or Workgroup Membership

- Determine the type of network access for which your computer will be configured
- You can specify a domain or workgroup through the Server Manager window
- Requirements for adding (joining) the computer to a domain:
  - Provide the DNS name of the domain you want to join
  - You must have an authorized user account in the domain you want to join
  - One domain controller and a DNS server must be online (reachable by the host) before you can join the domain

Using Server Manager for the Initial Setup

- Use Server Manager to begin configuring the server through tasks such as:
  - Setting the correct time and time zone
  - Configuring the Internet Protocol (IP) on the server's network interfaces
  - Configuring the firewall
  - Changing the default computer name and domain membership
  - Installing a modern Web browser
  - Activating the Windows Server operating system
  - Configuring server services and monitoring those services
  - Installing and managing server roles and program features
- You don't need to complete all the tasks immediately
  - But you should complete some preliminary tasks right away

Choosing a Computer Name

- The installation process assigns a randomly generated name for the server computer
- Some organizations have a predetermined naming scheme for computers on their network
- Microsoft's recommendations for creating a computer name include the following:
  - The maximum length is 63 characters
  - Use shorter names up to 15 characters (to reduce complexity, backwards compatibility with NetBIOS)
  - The computer name is unique in the Domain and network
  - If no DNS server exists on the network, use only standard Internet characters
  - If a DNS server is present on the network, use standard Internet characters plus additional characters such as $, %, &,* , and others

Setting the Time Zone

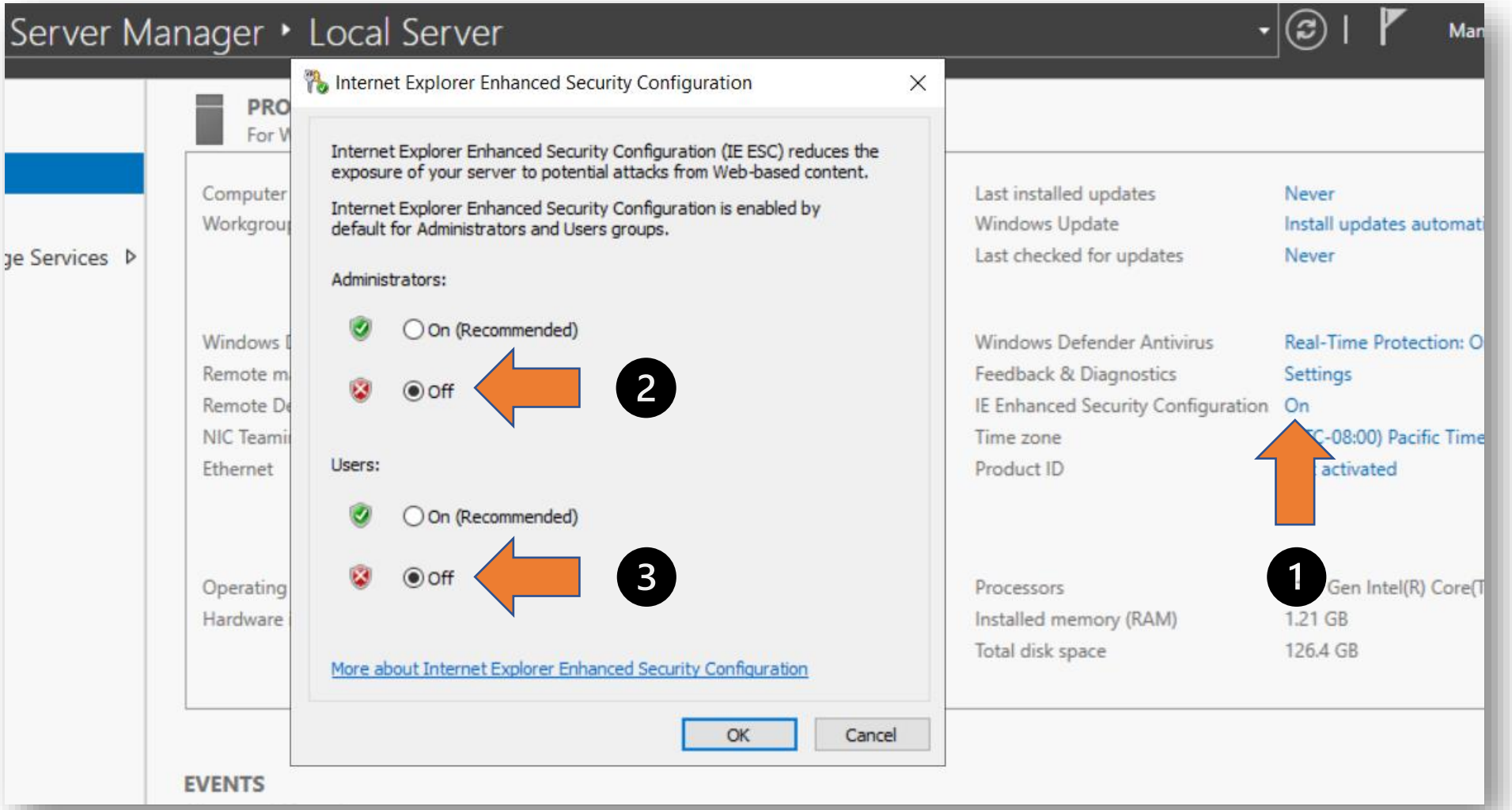
- Time is *very* important when providing Services – and securing those services – on a network and in a Domain
- If the clock differs by more than five minutes to that of the Domain Controller, the computer will not be able to join, communicate, or function within the domain.
- Time deviance can also affect security protocols, such as TLS/SSL.
- Time should be set by Time Zone. This will allow Windows to automatically synchronize with an external Network Time Protocol (NTP) Server.
- All domain members will use the Domain Controller as their NTP server

Installing a Web Browser

- Server 2019 does not use Edge. It uses Internet Explorer.
- Server 2022 *does* use Edge.
- Internet Explorer is hamstrung by a setting in Server Called Internet explorer enhanced Security Configuration (IE ESC).

  (*Please note, it is unusual for a Server to be used for browsing the internet. This is why IE is locked down. Usually, if a file is required on a server, it is downloaded elsewhere and then transferred to the server*)
- To download a modern browser (if need) you must first turn off IE ESC to access the internet

Turning off IE ESC

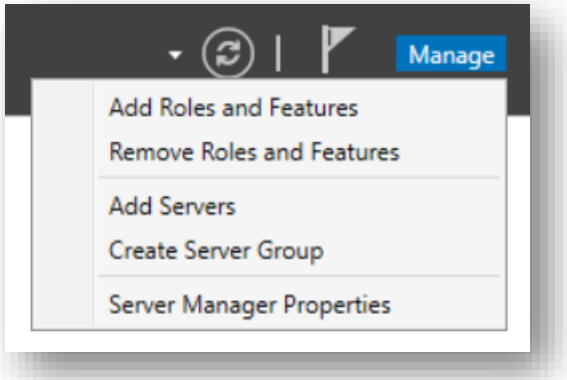


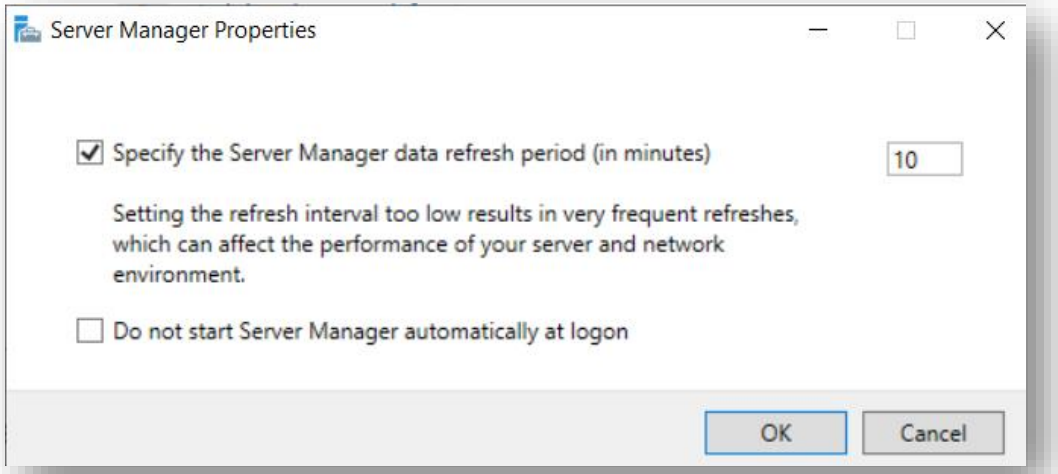
Anatomy of Server Manager

- Dashboard
  - Shows Server Roles and any events (from even viewer), services, and performance data relevant to those roles.
  - You can also run a Best Practice Analyzer (BPA) from here.
  - The Dashboard refreshes every 10 minutes. Used the refresh button at the top to do this manually.
- Notification Area
  - When tasks are started or finished or require your attention, this flag will have an indicator (or a number) on it.

Anatomy of Server Manager

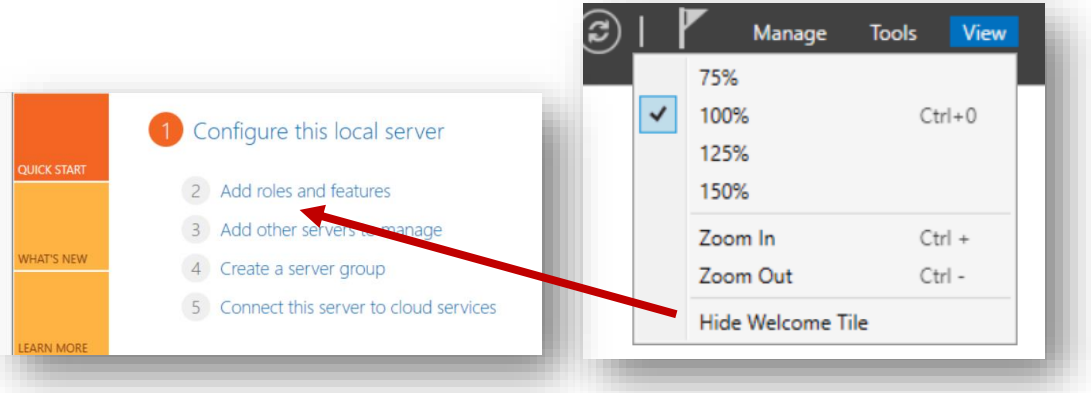
- Manage
  - Add Roles and Feature
  - Remove Roles and Features
  - Add Servers: Add servers to this console for management
  - Create Server groups: used for bulk configuration tasks.
  - Server Manager Properties: configure the refresh interval of the Dashboard and enable/disable Server Manager auto-launch at logon.

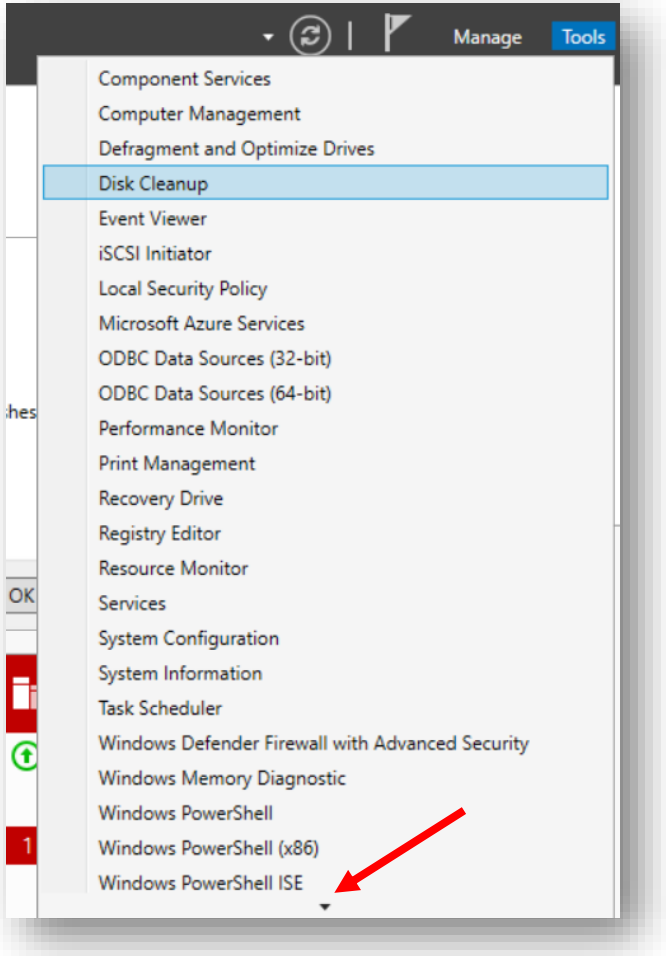




Anatomy of Server Manager

- **Tools**
  - Access most MMC snap-ins and other tools from here.
  - Note the down arrow at the bottom if the list is truncated.
- **View**:
  - Hide or show the welcome Tile





Roles, Role Services, and Features

Server Roles

- a set of software programs that, when installed and configured, lets a computer perform a specific function for multiple users or other computers within a network.
- describe the primary function, purpose, or use of a computer.
- provide users throughout an organization access to resources managed by other computers
- can queue user or computer requests, or record information about network users and computers that relates to the role.
- roles function automatically.

Roles, Role Services, and Features

Role services

- Role services are software programs that provide the functionality of a role.
- When you install a role, you can choose which role services the role provides for other users and computers in your enterprise.
- Some roles (e.g. DNS) have only a single function, and therefore do not have available role services.
- Other roles (RDP Services, Hyper-V) have several role services that can be installed, depending on the remote computing needs of your enterprise.
- a role can be one or a grouping of closely related, complementary role services (installing the role means installing one or more of its role services)

Roles, Role Services, and Features

Features

- Features are software programs that can
  - support or augment the functionality of one or more roles
  - improve the functionality of the server

Installing Roles and Features

- Read: Book Module 2, page 73-79 for a refresh on installing Roles and Features in using Server Manager

Using the Best Practices Analyzer (BPA) for Server Roles

- Once you've installed a role and set it up
  - It is important to determine that you have followed the best practices for the establishment of the role
- You can run the BPA to determine if one or more roles are installed and configured to follow the guidelines recommended by Microsoft
- BPA can be run against the server itself, or for each individual role.
- When problems are found, the analysis of each role yields a report that shows three levels of severity:
  - Information
  - Warning
  - Error

Using the Best Practices Analyzer (BPA) for Server Roles

- The guidelines the BPA uses for analysis are:
  - Configuration
  - Pre-deployment
  - Post-deployment
  - Performance
  - BPA Prerequisites
- General steps for running BPA:
  - Open Server Manager
  - Click Local Server in the left pane
  - In the right pane, scroll to Best Practices Analyzer
  - Click the down arrow for TASKS
  - Click Start BPA Scan and wait for the scan to complete
  - Read the results for the scan

Using Server Manager for the Initial Setup (cont'd)

- Many server administrators check for the following:
  - Remote management is enabled (Local Server)
  - Windows Firewall is working (mainly to allow RDP), or they disable it to set up other firewall software
  - Windows Update is configured to ensure latest updates are obtained
  - Static IP address are configured on Servers and Infrastructure devices.
- In the LAB:
  - Disable IPv6 on your NIC(s)
  - Turn off the firewall
  - Use the MS/Cisco Pa$$w0rd

Client Access Licenses

Planning User and Device Licensing

Core-based/Processor-based Licensing (The OS) VS. CALs

- Access for a user or device is enabled through a client access licenses (CALs)
  - One CAL is a license for one client or device to access one or more Windows servers on a network
- Licensing for the OS itself is done through ***physical*** processor cores
  - *See licensing Datasheet in BS*
- Standard and Datacenter editions require both CALs and core-based licensing
- Essentials Editions use processor-based licensing only.

Core Licensing

- Only Physical cores are counted when determining Cor licensing requirements.
  - Logical cores – hyperthreading, etc. – are not counted.
- Sold in 2-core packs.
  - Server has one processor w/2 cores → 16 cores licenses required…
  - Server has two processors each w/10 cores → 20 core licenses required.
  - Server has two processors each w/12 cores → 24 core licenses required.

User/Device Licensing (CALs)

- CALs only need to be purchased **one per version** (2019, 2022) of Server.
  - If you are running different versions of server, you will need to buy the CALs *twice.*
- User/Device CAL
  - Licenses one *device* **or** one *user* accessing *any* software running on any computer running Windows Server.
- External Connector
  - Used to allow *customers* to connect to services inside your network.
  - Cannot be used to license employees, contractors, or affiliates.

https://www.microsoft.com/en-us/licensing/product-licensing/client-access-license?oneroute=true